

Crop Disease Detection Under Realistic Environmental Distortions Using Synthetic Water Droplet Augmentation

Aarti Bandgar, School of Computer Science, University of Galway, H91 TK33 Galway, Ireland

Supervised by

Dr Priyanka Verma, School of Computer Science, University of Galway, H91 TK33 Galway, Ireland

Abstract—Plant disease detection models trained on clean images loss significant accuracy when tested on wet leaf images in field conditions. This paper introduces a physics-inspired synthetic water droplet augmentation pipeline that renders realistic droplets onto clean PlantVillage leaf images, creating a controlled distorted dataset for robustness evaluation. Three lightweight Convolutional Neural Network (CNN) architectures: MobileNetV2, EfficientNetB0 and ShuffleNetV2 are trained and evaluated under two settings: clean images (Baseline) and synthetic droplet images (Distorted). All experiments use the full PlantVillage dataset (54,305 images, 38 disease classes). A cross-dataset robustness test reveals that baseline models, trained only on clean images, drop in accuracy by 31-41% points when tested on wet leaf images: MobileNetV2 falls from 99.59% to 63.54%, EfficientNetB0 from 99.77% to 59.02%, and ShuffleNetV2 from 99.61% to 67.94%. Models trained directly on distorted images maintain accuracy above 99% on distorted test sets, and Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations confirm they focus on disease regions rather than droplet regions. These results demonstrate that water droplets are a genuine threat to field deployment and that training on synthetic distorted data is an effective response.

Index Terms—Crop disease detection, synthetic water droplet augmentation, transfer learning, PlantVillage, Deep Learning

I. INTRODUCTION

Plant disease cause major crop losses worldwide and early detection is essential for food security [1], [2]. Deep learning, particularly CNNs, has made automated disease diagnosis from smartphone photographs practical, with several studies reporting accuracy above 99% on PlantVillage benchmark [3], [4].

Crop diseases such as leaf damage, rust can wipe out a significant portion of a harvest if not caught early. In developing countries where many farmers cannot afford laboratory testing, a reliable smartphone-based tool could make a real difference. The idea is simple: a farmer takes a photo of a sick leaf, the app runs a model on the phone, and within seconds the farmer knows what disease is present and what treatment to apply. This works very well when images are clear and taken in good light.

Despite these results, a well-known problem remains. Mohanty et al. [3] showed that models trained on clean laboratory images fall from 99.35% accuracy to roughly 31% on real field photographs. One cause that has received little attention is

water: leaves are frequently wet after rain or dew and droplets partially obscure disease spots, change the leaf appearance and confuse models that have only seen dry leaves. No published study has directly measured this effect or used a controlled distortion pipeline to study it.

To understand how serious this problem is, consider what happens in practice. A farmer is most likely to check their crops for disease during or right after rain, which is exactly when leaves are wet. If the detection model only works on dry leaves, it is essentially useless at the moment when the farmer needs it most. This is the everyday reality of farming in most parts of the world, yet no study has built a controlled experiment around this problem.

The challenge in studying this is that no public dataset of wet diseased leaves exists. Collecting thousands of labelled wet leaf photographs would take months and significant resources. Instead, this study synthesizes realistic water droplets on clean leaf images using a physics-inspired rendering pipeline. This produces a controlled, reproducible distorted dataset for training and evaluating models under simulated wet conditions.

This paper addresses that gap. A synthetic droplet generation pipeline is built that adds realistic water droplets to clean PlantVillage images. Three lightweight models- MobileNetV2 [5], EfficientNetB0, and ShuffleNetV2 are trained on both clean and distorted data and then cross tested across datasets to measure how much accuracy drops on wet leaves and whether distortions aware training recovers it.

The specific contributions are:

- 1) A synthetic water droplet generation pipeline is designed and implemented that places realistic droplets on leaf images using refraction, Gaussian blur, specular highlights, and alpha blending. The pipeline is fully parameterized and reproducible, and it produces a one-to-one distorted copy of the entire PlantVillage dataset.
- 2) A cross-dataset robustness test is conducted in which models trained only on clean images are evaluated on the distorted test set. This directly measures, for the first time, how much accuracy a standard disease detection model loses when leaves are wet. The results show drops of 31 to 41 percentage points across all three architectures.

- 3) Three lightweight CNN architectures are trained and compared under both clean and distorted conditions: MobileNetV2 [5], EfficientNetB0 [6], and ShuffleNetV2 [7]. These models are chosen because they are designed for mobile deployment and can run on a smartphone without needing a powerful GPU.

II. LITERATURE REVIEW

A. Deep Learning for Crop Disease Detection

The foundational work in this field is by Mohanty et al. [3], who trained AlexNet and GoogLeNet on PlantVillage and achieved 99.35% accuracy on clean test images. However, the same paper reported a sharp drop to around 31% on real field photographs. This is one of the clearest early demonstrations of the gap between laboratory and field accuracy, and it is the central motivation for this study.

Pacal et al. [4] reviewed 160 disease detection studies from 2020 to 2024 and found that while high accuracy is now routine on benchmark datasets, environmental variability and limited data diversity remain unsolved problems. They specifically named synthetic data generation as an underused but promising solution, which directly supports the direction of this project.

Shoaib et al. [8] confirmed that augmentation techniques simulating diverse conditions can improve model robustness, though none of the reviewed work addressed water droplets specifically. Kamilaris et al. [9] surveyed 40 CNN-based agricultural studies and confirmed that limited and unrepresentative datasets remain the key obstacle to real-world deployment.

Together, these works show that the field achieves excellent results on clean images but has not addressed what happens when imaging conditions differ from training data.

B. Water Droplet and Rain Distortion Modeling

Shao et al. [10] proposed a multi-scale attention network for raindrop removal from single photographs and clearly described how droplets corrupt images through central occlusion and edge blurring. Wei et al. [11] introduced Derain-CycleGAN, which can both remove and synthesise rain on general outdoor images using unpaired training. Both works are relevant here because they characterize the visual effects this pipeline must replicate. Neither was applied to leaves or to disease classification.

C. Synthetic Augmentation in Agriculture

Lu et al. [12] reviewed GAN-based augmentation (DCGAN, CycleGAN, StyleGAN) in agriculture and found it can enlarge unbalanced datasets and improve model accuracy. However, GAN-based methods do not give per-pixel control over distortion location, which is needed for a targeted study of droplet effects. Krishna et al. [13] improved generalization by fine-tuning EfficientNet-B3 on a combined PlantDoc and web-sourced dataset, reaching 80.19% on field images. Their approach relied on data diversity rather than controlled distortion modeling.

D. Agricultural AI and Real-World Deployment

The motivation for building robust crop disease detection systems goes beyond academic benchmarks. Fan et al. [1] reviewed how AI contributes to the United Nations Sustainable Development Goals and found that AI has the potential to support 134 of the 169 SDG targets, with food security (SDG 2) being one of the most directly addressable areas. Automated disease diagnosis is one practical way this contribution can happen, as it gives smallholder farmers access to expert-level diagnosis without needing specialist equipment or a trained agronomist on site.

Goralski [2] examined the PlantVillage project specifically and reported the same laboratory to field accuracy collapse observed by Mohanty et al. [3], noting that classifiers which achieve close to 99% accuracy on high-quality studio photographs fall to around 32% on real photographs taken by farmers in the field. They also highlighted the risk of misdiagnosis when disease symptoms look visually similar, which can lead a farmer to apply the wrong treatment. These are not theoretical concerns they are practical failures that affect real farming decisions.

Both papers together make the same point: a crop disease detection system that only works under perfect conditions is not truly useful. For AI to deliver on its promise of supporting food security, the models must be reliable in the conditions farmers actually face, which includes wet leaves, uneven lighting, and photographs taken quickly in the field rather than carefully in a laboratory. This study directly targets one of those real-world conditions such as water droplets on leaf surfaces that has not been addressed in the existing literature.

E. Research Gap

The plant disease literature achieves high accuracy on clean datasets but does not model wet leaf conditions. The computer vision literature models water distortion accurately but has never applied it to disease classification. Two gaps result: no distorted leaf dataset exists and no study has measured the accuracy impact of water droplets on disease detection models specifically. This paper fills both.

III. PROPOSED APPROACH

The proposed approach produces a synthetic distorted version of the PlantVillage dataset and uses it to train and cross-test three lightweight disease detection models, as illustrated in Fig. 1. The pipeline takes clean leaf images, applies physics-inspired droplet rendering, and produces a one-to-one distorted counterpart for every image. Both datasets are then used under two experimental settings, as shown in Table I.

Three models are evaluated under both settings. MobileNetV2 [5] is the primary model, selected because it is designed for mobile deployment and matches the use case of a farmer using a smartphone in the field. EfficientNetB0 is included because Krishna et al. [13], the closest related paper, used EfficientNet architectures. ShuffleNetV2 is included as a lighter alternative (1.29M parameters) that has not been

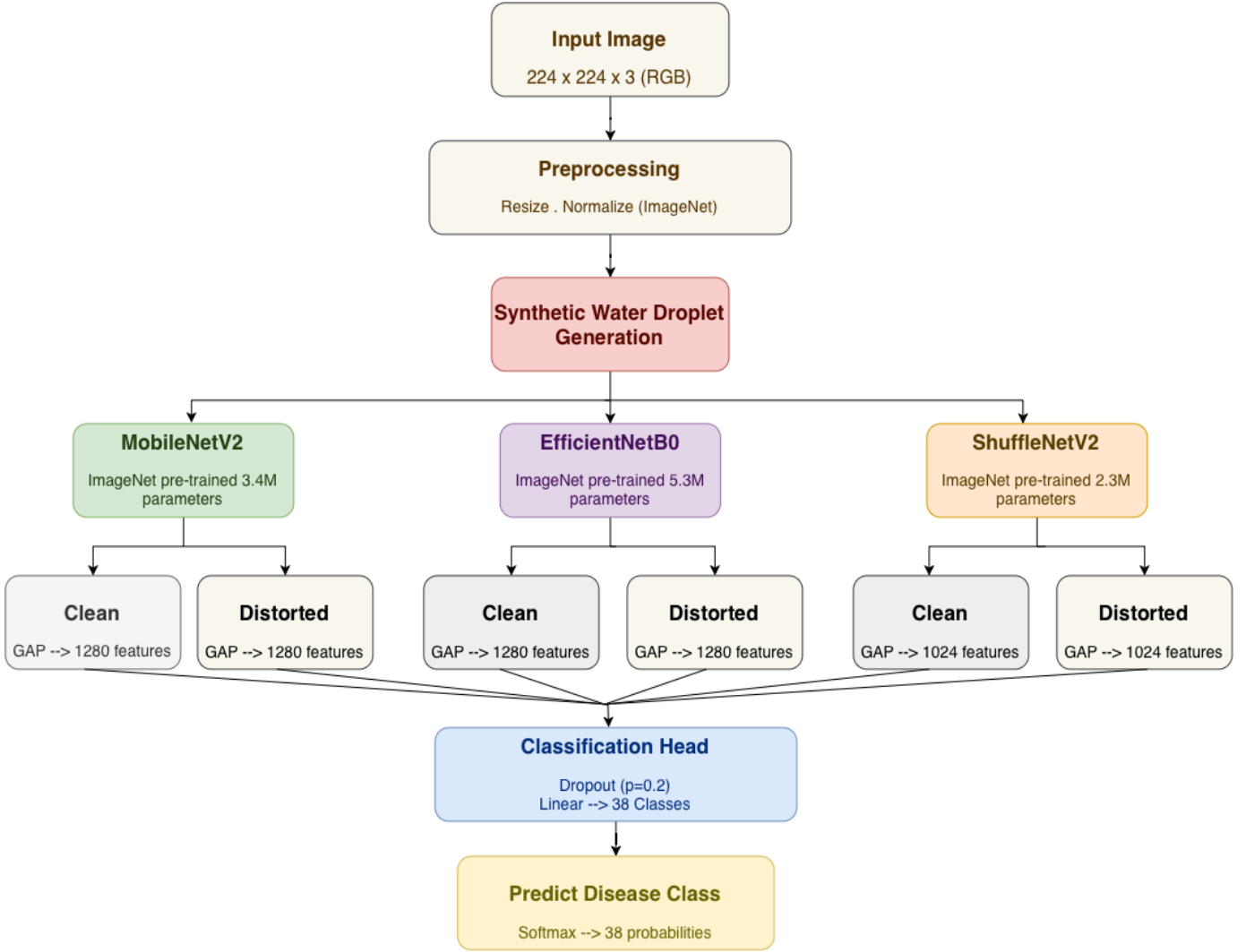


Fig. 1. Proposed models architecture pipeline.

TABLE I
EXPERIMENTAL SETTINGS

Setting	Training data	Purpose
Baseline	Clean only	Reference accuracy; cross-tested on droplets to measure robustness gap.
Distorted	Droplet only	Accuracy under wet conditions; cross-tested on clean for comparison.

previously evaluated on PlantVillage, representing a novel contribution to the comparison.

A. Synthetic Water Droplet Generation

The pipeline renders droplets exclusively within the detected leaf region, using HSV colour thresholding and morphological operations to create a binary leaf mask. Each droplet is

rendered through four sequential operations: (1) refraction, implemented via coordinate remapping, shifts the pixel content inside the droplet to simulate light bending through a curved water surface (2) Gaussian blur softens the leaf details under the droplet (3) a specular highlight reproduces the bright spot seen on real water droplets (4) alpha blending merges the rendered droplet with the original image at a controlled transparency. Droplet placement is constrained to leaf pixels, with parameters set to 22 droplets per image, radius 5 to 35 pixels, and blending transparency 0.60 to 0.85. These values were chosen through visual inspection to produce realistic distortions without fully hiding the leaf. Algorithm 1 summarizes the droplet generation process.

B. Model Architectures

All three models use ImageNet pretrained weights and the same classification head: Dropout($p=0.2$) followed by a fully connected layer mapping to 38 disease classes. Table II summarizes the three architectures.

Algorithm 1 Synthetic water droplet generation**Require:** Clean RGB leaf image I , size $224 \times 224 \times 3$ **Ensure:** Distorted image I'

- 1: Detect leaf region via HSV thresholding $\rightarrow$ mask M
- 2: Set: 22 droplets, radius $\in [5, 35]$ px, $\alpha \in [0.60, 0.85]$
- 3: **for** each droplet i **do**
- 4: Sample position (x, y) from leaf pixels in M
- 5: Apply refraction via coordinate remapping
- 6: Apply Gaussian blur (kernel 15×15)
- 7: Add specular highlight
- 8: Blend: $I' \leftarrow \alpha \cdot \text{droplet} + (1 - \alpha) \cdot I$
- 9: **end for**
- 10: **return** I'

TABLE II
MODEL ARCHITECTURES USED IN THIS STUDY

Model	Params	Feature dim	Selection reason
MobileNetV2	2.27M	1280	Mobile deployment, primary model
EfficientNetB0	4.06M	1280	Used in Krishna et al. [13]
ShuffleNetV2	1.29M	1024	Lightest model, novel for PlantVillage

C. Training Procedure

The same training procedure is used for all three models and both settings. Algorithm 2 gives the complete steps.

Algorithm 2 Training pipeline**Require:** Dataset D , ImageNet pretrained weights W **Ensure:** Trained model with best validation accuracy

- 1: Load model with weights W ; replace final layer $\rightarrow$ Linear($d, 38$)
- 2: Compute class weights (balanced); cap at 3.0
- 3: Adam optimizer, lr = 0.0001, weight decay = 10^{-4}
- 4: ReduceLROnPlateau: factor = 0.5, patience = 3
- 5: Split: Train 64%, Val 16%, Test 20% (stratified, seed 42)
- 6: **for** epoch = 1 to 20 **do**
- 7: Train with augmentation (flip, rotate $\pm 15^\circ$, jitter)
- 8: Evaluate on validation set
- 9: **if** best validation accuracy **then** save weights
- 10: **end if**
- 11: **if** no improvement for 5 epochs **then** stop; restore best
- 12: **end if**
- 13: **if** no improvement for 3 epochs **then** half the learning rate
- 14: **end if**
- 15: **end for**
- 16: **return** best model weights

Key training decisions: Weighted loss assigns higher penalty to minority classes, preventing 0% recall on rare diseases. Low learning rate (0.0001) preserves the useful ImageNet features while adapting the model to leaf diseases. Early stopping prevents overfitting and saves training time.

IV. EXPERIMENTAL ANALYSIS

A. Experimental Setup

All experiments were run on an Apple Silicon Mac (M-series chip) using the MPS (Metal Performance Shaders) GPU backend. The implementation uses PyTorch and torchvision for model training, scikit-learn for metrics and stratified splitting, OpenCV and NumPy for image processing, and matplotlib for visualization. All code is organized into a modular pipeline with separate modules for data loading, model building, training, and evaluation. Two experimental settings are evaluated (Baseline and Distorted, Table I) across three models (Table II), giving six training runs in total. A cross-dataset robustness test is then run in which each baseline model (trained on clean images) is evaluated on the distorted test set, directly measuring how much water droplets harm accuracy. The source code, experimental implementation, and configuration files are publicly available in the project repository.¹

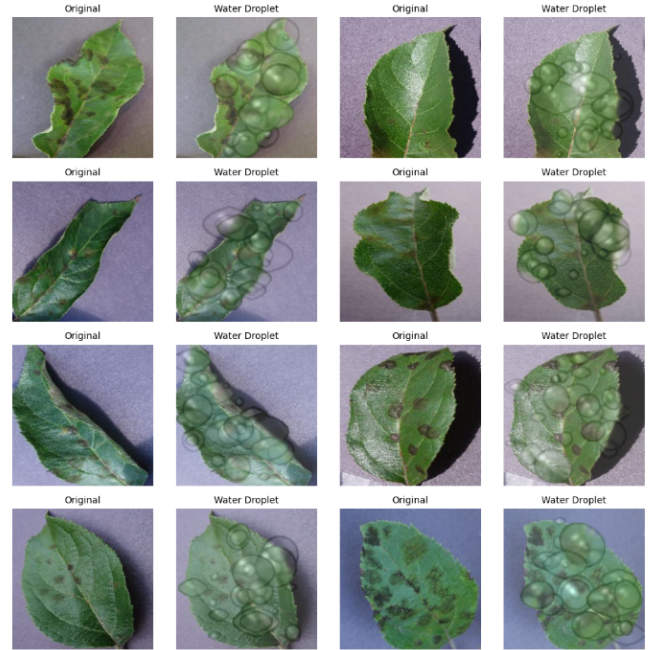
B. Datasets

Fig. 2. Sample clean and synthetic droplet images from PlantVillage.

PlantVillage Dataset: The study uses the PlantVillage dataset [3], [14], accessed via Kaggle², containing 54,305 colour leaf images across 38 disease classes. Images are resized to 224×224 pixels and normalized using ImageNet mean and standard deviation. The dataset is split into 64% train, 16% validation, and 20% test using a stratified split (seed 42) so that all classes are proportionally represented in each split. The distorted dataset is produced by applying

¹Source code: <https://github.com/Aarti0105/Crop-disease-detection>

²<https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>

TABLE III
TEST SET PERFORMANCE ACROSS ALL MODELS AND SETTINGS

Metric	MobileNetV2		EfficientNetB0		ShuffleNetV2	
	Baseline	Distorted	Baseline	Distorted	Baseline	Distorted
Accuracy (%)	99.59	99.37	99.77	99.55	99.61	99.12
Macro Precision (%)	99.26	99.17	99.69	99.10	99.33	98.82
Macro Recall (%)	99.51	99.33	99.70	99.39	99.48	99.02
Macro F1-Score (%)	99.38	99.24	99.70	99.24	99.40	98.91
Weighted F1-Score (%)	99.60	99.37	99.77	99.55	99.61	99.12

the droplet pipeline to every PlantVillage image, creating a one-to-one distorted copy with the same labels and splits. The same test set indices are used for both datasets, ensuring the cross-dataset evaluation is directly comparable.

Distorted Dataset: The distorted dataset is produced by applying the synthetic water droplet pipeline to every image in the PlantVillage dataset. This creates a one-to-one distorted version of the original image. The same class labels and train/val/test splits are applied. Sample images from both the clean and distorted datasets are shown in Figure 2

C. Results and Analysis

1) Training Performance: All six models converged quickly during training, which is expected given that all three architectures start from ImageNet pretrained weights. For the Baseline setting, all three models reached above 97% validation accuracy by the end of the second epoch. The Baseline models converged slightly faster than the Distorted models, which makes sense because clean images are simpler to classify. The Distorted models took a few more epochs to reach the same level but still converged well and showed no signs of overfitting. In both settings and across all three models, the training and validation accuracy stayed close throughout training, which confirms that the models were learning genuine features and not memorizing the training data.

The reason validation accuracy is sometimes slightly higher than training accuracy in the early epochs is because training images receive random augmentation on every pass which are flips, rotations, colour changes which makes them harder to classify. Validation images are passed through without any augmentation, so they are more consistent, which leads to higher measured accuracy.

2) Test Set Results: Table III shows the test accuracy and overall metrics for all six model setting combinations, measured on the 20% test set held out completely during training.

Accuracy is the percentage of images the model got right. Precision measures, out of all images predicted as a particular disease, how many actually had that disease. Recall measures, out of all images that actually had a disease, how many the model correctly identified. F1-score combines precision and

recall into a single number, which is useful when class sizes are uneven. Two versions are reported: macro-averaged (all 38 classes weighted equally) and weighted-averaged (weighted by class size).

EfficientNetB0 achieves the highest accuracy in both settings: 99.77% on clean images and 99.55% on distorted images. MobileNetV2 and ShuffleNetV2 follow closely, both above 99.59% on clean images and above 99.12% on distorted images.

The accuracy difference between the Baseline and Distorted settings is small for all three models: 0.22 percentage points for EfficientNetB0, 0.22 for MobileNetV2, and 0.49 for ShuffleNetV2. This shows that when a model is trained on distorted images, it can still classify diseases very accurately even when leaves are wet. ShuffleNetV2 shows the largest gap, which is likely because it is the smallest model and has less capacity to handle the added complexity of the droplets. Even so, it still achieves above 99% on clean images, which is strong for a model with only 1.29 million parameters.

TABLE IV
CROSS-DATASET ROBUSTNESS: CLEAN-TRAINED MODELS ON DROPLET IMAGES

Model	On clean	On droplet	Drop
MobileNetV2	99.59%	63.54%	−36.06%
EfficientNetB0	99.77%	59.02%	−40.75%
ShuffleNetV2	99.61%	67.94%	−31.67%

3) Cross-Dataset Robustness Test: This is the most important experiment in the study. Each baseline model trained only on clean images and evaluated on the distorted test set. This directly simulates what happens if a farmer uses a standard, clean-trained model on a day when the leaves are wet.

The results in Table IV are striking. All three models lose between 31 and 41 percentage points of accuracy when tested on wet leaf images they were not trained on. EfficientNetB0, which performs best on clean images, also drops the most: from 99.77% down to 59.02%, meaning roughly 4 out of every 10 wet leaf images would be diagnosed incorrectly.

In absolute terms, MobileNetV2 makes approximately 44 wrong predictions on the clean test set. On the distorted test set, that number rises to around 3,950 roughly 90 times more errors, caused entirely by water droplets on the leaves.

TABLE V
CONFUSION MATRIX SUMMARY: ALL MODELS AND SETTINGS

Model	Setting	TP	FP	FN	TN
MobileNetV2	Baseline	10,815	46	46	401,811
MobileNetV2	Distorted	10,791	70	70	401,787
EfficientNetB0	Baseline	10,834	27	27	401,830
EfficientNetB0	Distorted	10,813	48	48	401,809
ShuffleNetV2	Baseline	10,817	44	44	401,813
ShuffleNetV2	Distorted	10,762	99	99	401,758

This directly answers the core question of the study. Water droplets are not a minor inconvenience. They cause a large and serious accuracy drop that would make these models unreliable in real field conditions. A model that is almost perfect in the lab can fail badly the moment it rains. The contrast with the same-distribution results in Table III makes the solution clear: training on synthetic wet leaf images restores accuracy above 99% for all three models.

4) *Confusion Matrix Analysis*: The confusion matrix breaks each model's predictions into four values: True Positives (TP) are images correctly classified, False Positives (FP) are images assigned to the wrong class, False Negatives (FN) are images that were missed, and True Negatives (TN) are correct rejections of other classes. It gives a more detailed picture of model behavior than accuracy alone, especially for identifying which types of errors are being made. All values in Table V are summed across all 38 disease classes.

Confusion Matrix

True Positive (TP) 10,838	False Positive (FP) 23
False Negative (FN) 23	True Negative (TN) 401,834

Fig. 3. MobileNetV2 Baseline: overall TP/FP/FN/TN (summed across all 38 classes).

MobileNetV2 (Figures 3 and 4): In the Baseline setting, the model makes 46 errors out of 10,861 test images. Training on distorted images increases this to 70 errors, which is a small rise and still a very strong result given that every test image contains water droplets.

Confusion Matrix

True Positive (TP) 10,816	False Positive (FP) 45
False Negative (FN) 45	True Negative (TN) 401,812

Fig. 4. MobileNetV2 Distorted: overall TP/FP/FN/TN (summed across all 38 classes).

Confusion Matrix

True Positive (TP) 10,836	False Positive (FP) 25
False Negative (FN) 25	True Negative (TN) 401,832

Fig. 5. EfficientNetB0 Baseline: overall TP/FP/FN/TN (summed across all 38 classes).

EfficientNetB0 (Figures 5 and 6): This model produces the fewest errors across both settings. The Baseline makes only 27 wrong predictions and the Distorted setting increases this to 48, the smallest jump of all three models, showing that EfficientNetB0 handles the droplet distortion with the least impact on precision.

Confusion Matrix	
True Positive (TP) 10,812	False Positive (FP) 49
False Negative (FN) 49	True Negative (TN) 401,808

Fig. 6. EfficientNetB0 Distorted: overall TP/FP/FN/TN (summed across all 38 classes).

ShuffleNetV2 (Figures 7 and 8): The Baseline makes 44 wrong predictions, which is similar to MobileNetV2 despite having fewer parameters. The Distorted setting shows the largest increase to 99 errors, reflecting the model’s smaller capacity to absorb the added visual complexity of the droplets. Even so, 99 wrong predictions out of 10,861 remains a strong result for the lightest model in the comparison.

The True Negative values exceed 400,000 in all six cases because TN is counted once per class per image. With 38 classes and over 10,000 test images, every correctly rejected class adds one to the total, making it large by design rather than by error.

Confusion Matrix	
True Positive (TP) 10,819	False Positive (FP) 42
False Negative (FN) 42	True Negative (TN) 401,815

Fig. 7. ShuffleNetV2 Baseline: overall TP/FP/FN/TN (summed across all 38 classes).

Confusion Matrix	
True Positive (TP) 10,765	False Positive (FP) 96
False Negative (FN) 96	True Negative (TN) 401,761

Fig. 8. ShuffleNetV2 Distorted: overall TP/FP/FN/TN (summed across all 38 classes).

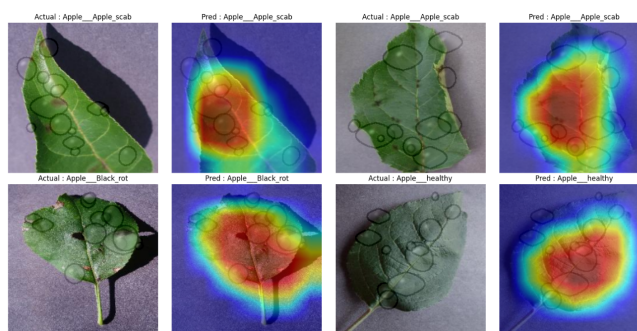


Fig. 9. Grad-CAM visualization for the MobileNetV2 Distorted model on wet leaf images.

5) *Grad-CAM Visualization*: Grad-CAM [18] was used to check whether the model is genuinely detecting disease or whether it is using the droplets themselves as a signal. It works by computing which parts of the image had the most influence on the model’s prediction and overlays the result as a heatmap: red and yellow for high attention, blue and green for low.

This is important for two reasons. First, it confirms that the model is genuinely identifying disease and not learning droplet patterns as a shortcut. Second, it shows that even with water droplets present, the model looks past them and finds the disease features underneath. This gives confidence that a model trained on synthetic wet images can be trusted in real field conditions. Figure 9 shows the Grad-CAM heatmaps for the distorted mobileNetV2 model applied to wet leaf images.

6) *Comparison with Existing Work*: Table VI compares this study against the five papers from the reference list that report actual disease classification accuracy numbers. The other references are review papers, rain and distortion papers, augmentation surveys, and architecture papers; they do not report crop disease accuracy and are not included.

TABLE VI
COMPARISON WITH EXISTING WORK ON CROP DISEASE DETECTION

Reference	Model	Dataset / Conditions	Accuracy	Year
<i>Clean PlantVillage images (standard benchmark)</i>				
Mohanty et al. [3]	AlexNet	PlantVillage: clean	99.35%	2016
Mohanty et al. [3]	GoogLeNet	PlantVillage: clean	99.53%	2016
Delnevo et al. [15]	MobileNetV2 + IoT	PlantVillage: clean	99.16%	2021
Garg et al. [16]	MobileNetV2 (CROPCARE)	PlantVillage: clean	96.12%	2021
Iqbal [17]	Transformer + CNN	PlantVillage: clean	96.20%	2025
This study: Efficient-NetB0	EfficientNetB0	PlantVillage: clean	99.77%	2026
<i>Real-world or distorted images</i>				
Krishna et al. [13]	EfficientNet-B3	PlantDoc + web: field	80.19%	2025
This study: Distorted models	MobileNetV2, Efficient-NetB0, ShuffleNetV2	PlantVillage: droplets	99.12 to 99.55%	2026
This study: Cross-dataset	All three (clean-trained on wet images)	Clean model: droplet test	59 to 68%	2026

On clean PlantVillage images, EfficientNetB0 at 99.77% matches or exceeds all prior results in the table, including studies that add IoT systems or transformer components on top of their base model.

The cross-dataset row in the table has no equivalent in any prior study. No existing paper has reported what happens when a clean-trained model encounters wet leaf images. This study measures it for the first time, revealing a drop to 59 to 68% which is the clearest evidence yet that clean-image benchmarks do not represent real field conditions. The distorted-trained model results of 99.12 to 99.55% show this problem has a direct solution.

V. CONCLUSION

This paper showed that water droplets are a serious and measurable threat to crop disease detection in real field conditions. All three models tested on MobileNetV2, EfficientNetB0, and ShuffleNetV2 which achieve above 99% accuracy on clean PlantVillage images, but drop to between 59% and 68% when the same clean-trained models are evaluated on wet leaf images. EfficientNetB0 drops by 40.75% points, MobileNetV2 by 36.06%, and ShuffleNetV2 by 31.67%. In practical terms, MobileNetV2 goes from 46 wrong predictions on clean images to approximately 3,950 on wet images: roughly 90 times more errors caused entirely by water on the leaves. When all three models are trained on synthetic distorted images instead, they recover to above 99% accuracy on distorted test sets, with EfficientNetB0 reaching 99.55%, MobileNetV2 99.37%, and ShuffleNetV2 99.12%. Grad-CAM

confirms these models focus on disease regions and not on droplet regions, so the recovery in accuracy is genuine.

The main contribution of this work is a reproducible synthetic water droplet pipeline that generates the first controlled wet-leaf disease dataset from PlantVillage and a cross-dataset evaluation framework that makes the accuracy impact directly measurable. Future work should validate these findings using real photographs of wet diseased leaves collected in the field, and could extend the pipeline to simulate other environmental conditions such as shadows, mud, and varying lighting.

VI. USE OF AI

During the preparation of this paper, AI-based tools were used as a supportive aid for initial brainstorming, structuring ideas, and outlining the manuscript. Additionally, these tools were utilized during the technical development phase to assist with code debugging, identifying potential implementation issues, and troubleshooting syntax or runtime errors. They were also used to refine sentence structure and improve language clarity.

The Grad-CAM visualization code was generated with AI assistance. This was used specifically to inspect which regions of the leaf image the model focused on when making its predictions, in order to verify that the model was attending to actual disease lesions rather than to the synthetic water droplet regions. Grad-CAM is not a main contribution of this work, it was included only as an additional qualitative analysis to support the interpretation of the model's behavior.

I have performed all of the major research decisions, including the methodological approach, selection of algorithms

and datasets, experimental design, implementation, evaluation, analysis, and interpretation of the findings.

REFERENCES

- [1] Z. Fan, Z. Yan, and S. Wen, "Deep learning and artificial intelligence in sustainability: a review of sdgs, renewable energy, and environmental health," *Sustainability*, vol. 15, no. 18, p. 13493, 2023.
- [2] M. A. Goralski and T. K. Tan, "Artificial intelligence and sustainable development," *The International Journal of Management Education*, vol. 18, no. 1, p. 100330, 2020.
- [3] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in plant science*, vol. 7, p. 1419, 2016.
- [4] I. Pacal, I. Kunduracioglu, M. H. Alma, M. Deveci, S. Kadry, J. Nedoma, V. Slany, and R. Martinek, "A systematic review of deep learning techniques for plant diseases," *Artificial Intelligence Review*, vol. 57, no. 11, p. 304, 2024.
- [5] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4510–4520, 2018.
- [6] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, pp. 6105–6114, PmlR, 2019.
- [7] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "Shufflenet v2: Practical guidelines for efficient cnn architecture design," in *European conference on computer vision*, pp. 122–138, Springer, 2018.
- [8] M. Shoaib, A. Sadeghi-Niaraki, F. Ali, I. Hussain, and S. Khalid, "Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions," *Frontiers in Plant Science*, vol. 16, p. 1538163, 2025.
- [9] A. Kamilaris and F. X. Prenafeta-Boldú, "A review of the use of convolutional neural networks in agriculture," *The Journal of Agricultural Science*, vol. 156, no. 3, pp. 312–322, 2018.
- [10] M.-W. Shao, L. Li, D.-Y. Meng, and W.-M. Zuo, "Uncertainty guided multi-scale attention network for raindrop removal from a single image," *IEEE Transactions on Image Processing*, vol. 30, pp. 4828–4839, 2021.
- [11] Y. Wei, Z. Zhang, Y. Wang, M. Xu, Y. Yang, S. Yan, and M. Wang, "Deraincyclegan: Rain attentive cyclegan for single image deraining and rainmaking," *IEEE Transactions on Image Processing*, vol. 30, pp. 4788–4801, 2021.
- [12] Y. Lu, D. Chen, E. Olaniyi, and Y. Huang, "Generative adversarial networks (gans) for image augmentation in agriculture: A systematic review," *Computers and Electronics in Agriculture*, vol. 200, p. 107208, 2022.
- [13] M. S. Krishna, P. Machado, R. I. Otuka, S. W. Yahaya, F. Neves dos Santos, and I. K. Ihianle, "Plant leaf disease detection using deep learning: A multi-dataset approach," *J*, vol. 8, no. 1, p. 4, 2025.
- [14] D. Hughes, M. Salathé, *et al.*, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015.
- [15] G. Delnevo, R. Girau, C. Ceccarini, and C. Prandi, "A deep learning and social iot approach for plants disease prediction toward a sustainable agriculture," *IEEE Internet of Things Journal*, vol. 9, no. 10, pp. 7243–7250, 2021.
- [16] G. Garg, S. Gupta, P. Mishra, A. Vidyarthi, A. Singh, and A. Ali, "Cropcare: An intelligent real-time sustainable iot system for crop disease detection using mobile vision," *IEEE Internet of Things Journal*, vol. 10, no. 4, pp. 2840–2851, 2021.
- [17] A. Iqbal, "Planthealthnet: Transformer-enhanced hybrid models for disease diagnosis and severity estimation in agriculture," *IEEE Access*, 2025.
- [18] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE international conference on computer vision*, pp. 618–626, 2017.